

CS6530 – Applied Cryptography

Jul–Nov 2026

Assignment 1

Secure Data Protection using AES-GCM and ChaCha20-Poly1305

1. Before You Begin

Welcome to the first programming assignment in CS6530 – Applied Cryptography.

The objective of this assignment is to apply modern authenticated encryption techniques to design, implement, test, and evaluate a secure data protection subsystem. The assignment emphasizes secure engineering practice rather than the implementation of cryptographic primitives.

You will develop a subsystem that protects generic chunked/packetized application data (application records) exchanged between a sender and a receiver using two modern Authenticated Encryption with Associated Data (AEAD) configurations:

- AES-GCM
- ChaCha20-Poly1305

The same subsystem shall support both configurations and demonstrate equivalent functionality.

This assignment intentionally uses generic chunked/packetized application data (application records) rather than a specific application domain. This allows you to focus on the engineering principles of secure data protection while producing a solution that is applicable to a wide range of practical systems, including telemetry, cloud services, media delivery, messaging systems, and communication networks.

This assignment assumes that the sender and receiver already share a secret cryptographic key before communication begins. Secure key establishment and key management are outside the scope of this assignment and will be addressed later in the course.

You are expected to use an appropriate cryptographic library for implementing the required functionality. Implementation of the AES, ChaCha20, GCM, or Poly1305 cryptographic primitives is not required. Instead, the emphasis is on correctly integrating these algorithms into a secure engineering solution.

Throughout this assignment, you are expected to demonstrate good engineering practice by:

- designing a well-structured secure data protection subsystem;
- selecting and documenting an appropriate nonce management approach;
- protecting application metadata using Associated Data (AAD);
- implementing an appropriate replay handling mechanism;
- systematically validating the implementation against the specified Testing Requirements; and
- comparing the performance of AES-GCM and ChaCha20-Poly1305 under equivalent experimental conditions.

Successful completion of this assignment requires not only a working implementation but also clear evidence that the subsystem satisfies the stated Functional Requirements and Security Requirements through systematic testing and analysis.

Students are encouraged to begin the assignment early and validate their implementation incrementally rather than waiting until the final stages of development.

2. Assignment Overview

The objective of this assignment is to design, implement, test, and evaluate a secure data protection subsystem for generic chunked/packetized application data (application records) using modern Authenticated Encryption with Associated Data (AEAD) techniques.

The subsystem shall support the following two AEAD configurations:

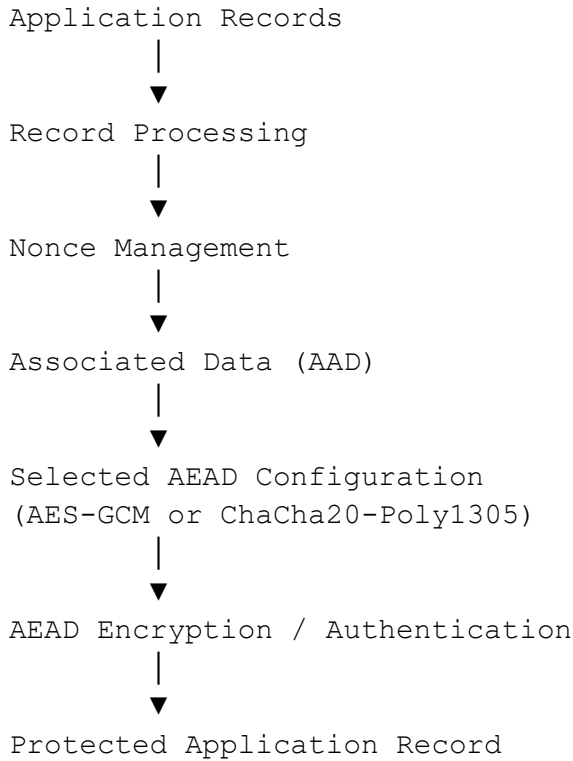
- AES-GCM
- ChaCha20-Poly1305

During a particular execution, the subsystem shall operate using the selected AEAD configuration (AES-GCM or ChaCha20-Poly1305). The same subsystem shall provide equivalent functionality when operated using either of the two supported AEAD configurations. All mandatory Testing Requirements shall subsequently be demonstrated separately using both configurations.

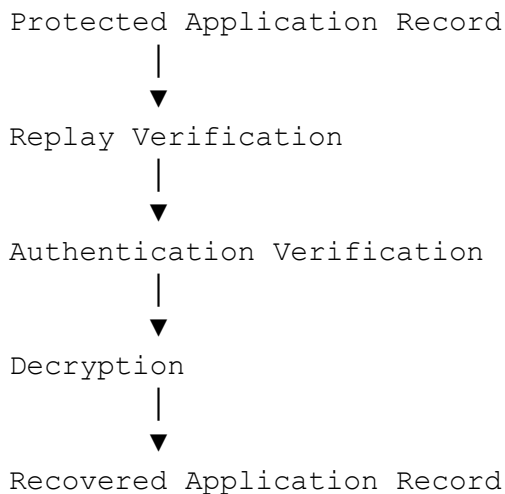
The assignment intentionally uses generic chunked/packetized application data (application records) rather than a specific application domain. This enables the engineering principles developed in this assignment to be readily applied to a variety of practical systems, including telemetry, messaging, cloud services, media delivery, communication networks, and other applications that exchange protected data records.

The primary objective of this assignment is not simply to invoke a cryptographic library to encrypt data. Instead, students are expected to engineer a complete secure data protection subsystem that correctly integrates authenticated encryption into the processing of application records.

A typical logical view of the subsystem is shown below.



At the receiving side, the protected application record shall be processed through verification before the original application record is recovered.



The assignment requires students to demonstrate correct engineering of the complete secure data protection subsystem, including nonce management, Associated Data (AAD), replay handling, validation against the specified Testing Requirements, and performance evaluation.

Upon successful completion of this assignment, students should be able to:

- engineer a secure data protection subsystem using modern AEAD techniques;

- correctly apply the selected AEAD configuration (AES-GCM or ChaCha20-Poly1305) for protecting application records;
- design appropriate mechanisms for nonce management, Associated Data (AAD), and replay handling;
- systematically validate the implementation against the specified Testing Requirements; and
- compare the performance characteristics of the two supported AEAD configurations under equivalent experimental conditions.

3. Assignment Scope and Assumptions

This assignment focuses on the engineering, implementation, testing, and evaluation of a secure data protection subsystem for generic chunked/packetized application data (application records) using modern Authenticated Encryption with Associated Data (AEAD) techniques.

The assignment is designed to provide practical experience in integrating cryptographic algorithms into an engineering solution. The emphasis is on the correct application of authenticated encryption rather than the implementation of the underlying cryptographic primitives.

3.1 Scope

The scope of this assignment includes:

- protection of generic chunked/packetized application data (application records);
- operation using the selected AEAD configuration (AES-GCM or ChaCha20-Poly1305);
- secure processing of application records between a sender and a receiver;
- nonce management;
- protection of Associated Data (AAD);
- implementation of an appropriate replay handling mechanism;
- validation against the specified Testing Requirements; and
- comparison of the performance of the two supported AEAD configurations.

3.2 Out of Scope

The following topics are outside the scope of this assignment:

- implementation of the AES, ChaCha20, GCM, or Poly1305 cryptographic primitives;
- secure key establishment and key exchange protocols;
- public key infrastructure (PKI) and certificate management;
- implementation of transport-layer or communication protocols;
- reliable packet delivery, packet ordering, or packet retransmission mechanisms; and
- network security mechanisms beyond the secure protection of application records.

3.3 Assumptions

For the purpose of this assignment, the following assumptions shall apply:

- The sender and receiver already share a common secret cryptographic key.
- Secure key establishment is outside the scope of this assignment.
- Any suitable cryptographic library may be used to implement the required functionality.
- Students are not required to implement the AES, ChaCha20, GCM, or Poly1305 cryptographic primitives.
- The sender, receiver, and malicious actor may be implemented as logical entities within a single application or as separate communicating programs.
- Network communication is optional. A logical exchange of application records between the sender and receiver is sufficient, provided that all Functional Requirements and Testing Requirements are satisfied.

Students are encouraged to clearly document the implementation assumptions made during the development of their solution in the Assignment 1 Report.

4. Functional Requirements

The secure data protection subsystem shall satisfy the following Functional Requirements.

FR-1 – Application Record Protection

The subsystem shall protect generic chunked/packetized application data (application records) exchanged between a sender and a receiver. Each application record shall be processed independently by the subsystem.

FR-2 – AEAD Configuration Support

The subsystem shall support operation using the selected AEAD configuration (AES-GCM or ChaCha20-Poly1305). The same functionality shall be available when either of the two supported AEAD configurations is selected.

FR-3 – Protected Application Record Generation

The subsystem shall generate a protected application record by applying authenticated encryption to an application record using the selected AEAD configuration.

FR-4 – Associated Data (AAD) Support

The subsystem shall support the inclusion of Associated Data (AAD) during authenticated encryption and authenticated decryption. The selected AAD shall participate in the authentication process and shall be verified during application record recovery.

FR-5 – Nonce Management

The subsystem shall correctly manage nonces in accordance with the requirements of the selected AEAD configuration. The nonce management approach adopted shall be documented in the Assignment 1 Report.

FR-6 – Application Record Recovery

The subsystem shall successfully recover the original application record when a valid protected application record, the correct cryptographic key, the correct nonce, and the corresponding Associated Data (AAD) are provided.

FR-7 – Authentication Failure Handling

The subsystem shall detect authentication failures and reject any protected application record that fails authentication verification. Under such conditions, the subsystem shall not release the recovered application record.

FR-8 – Replay Handling

The subsystem shall implement an appropriate mechanism to detect and handle replayed protected application records. The replay handling strategy adopted shall be documented in the Assignment 1 Report.

FR-9 – Configuration Equivalence

The subsystem shall provide equivalent functionality when operated using either of the two supported AEAD configurations (AES-GCM or ChaCha20-Poly1305). The selected AEAD configuration shall not alter the functional behaviour of the subsystem beyond the cryptographic algorithm being used.

FR-10 – Engineering Documentation

The implementation shall be accompanied by appropriate documentation describing the subsystem design, nonce management approach, Associated Data (AAD) selection, replay handling strategy, implementation assumptions, and validation against the specified Testing Requirements as part of the Assignment 1 Report.

5. Security Requirements

The secure data protection subsystem shall satisfy the following Security Requirements.

SR-1 – Confidentiality Protection

The subsystem shall protect the confidentiality of application records by ensuring that application data is not disclosed in plaintext during the exchange of protected application records.

SR-2 – Integrity Verification

The subsystem shall ensure that unauthorized modification of a protected application record is identified during authentication verification. A protected application record that fails authentication verification shall be rejected by the receiver.

SR-3 – Nonce Management

The subsystem shall correctly manage nonces in accordance with the requirements of the selected AEAD configuration. During normal operation, the nonce management approach shall prevent reuse of a nonce with the same cryptographic key.

SR-4 – Associated Data (AAD) Protection

The subsystem shall support the authentication of Associated Data (AAD). Any modification of the Associated Data shall be detected during authentication verification.

SR-5 – Replay Protection

The subsystem shall implement an appropriate mechanism to detect and handle replayed protected application records in accordance with the documented replay handling strategy.

SR-6 – Authentication Failure Handling

The subsystem shall reject any protected application record that fails authentication verification. Under such conditions, the subsystem shall not release the recovered application record and shall indicate that authentication verification has failed.

The Security Requirements specified in this section shall be validated through the corresponding Testing Requirements described later in this assignment.

6. Testing Requirements

The implementation shall be validated against the following Testing Requirements.

Unless otherwise specified, every Testing Requirement (TR-1 to TR-8) shall be demonstrated separately using both supported AEAD configurations (AES-GCM and ChaCha20-Poly1305). Demonstrating a Testing Requirement using only one of the two supported AEAD configurations will receive only partial credit for that assessment component.

For the purpose of testing, the implementation shall model three logical entities:

- Sender – Generates valid protected application records.
- Malicious Actor – Generates or modifies protected application records to simulate invalid or malicious conditions.
- Receiver – Validates received protected application records and performs the appropriate processing.

The implementation may realize these entities as separate communicating programs or as logical components within a single application.

TR-1 – Positive Baseline Test

Demonstrate successful protection, transmission, verification, and recovery of valid application records between the sender and the receiver.

TR-2 – Ciphertext Integrity Test

Demonstrate that modification of the protected application record by the malicious actor results in authentication verification failure and rejection of the modified record by the receiver.

TR-3 – Authentication Tag Test

Demonstrate that modification of the authentication tag results in authentication verification failure and rejection of the protected application record.

TR-4 – Associated Data (AAD) Test

Demonstrate that modification of the Associated Data (AAD) results in authentication verification failure and rejection of the protected application record.

TR-5 – Replay Test

Demonstrate that replay of a previously accepted protected application record is detected and handled according to the documented replay handling strategy.

TR-6 – Wrong-Key Test

Demonstrate that use of an incorrect cryptographic key results in authentication verification failure and rejection of the protected application record.

TR-7 – Nonce Management Verification

Demonstrate that the nonce management approach satisfies the requirements of the selected AEAD configuration. The demonstration shall include evidence that nonce reuse does not occur during normal operation.

TR-8 – Performance Evaluation

Compare the performance of the two supported AEAD configurations by measuring the protection and recovery of application records of different sizes. As a minimum, performance measurements shall be obtained for 64 Bytes, 1 KiB and 64 KiB application records. Students shall compare the observed performance characteristics and briefly discuss their observations in the Assignment 1 Report.

Note: As a guideline, students are encouraged to demonstrate nonce management by processing approximately 10,000 application records using the same cryptographic key. Alternative approaches that clearly demonstrate correct nonce management are also acceptable.

Test Documentation

For every Testing Requirement (TR-1 to TR-8), the Assignment 1 Report shall include:

- Objective
- Procedure
- Test Input

- Expected Behaviour
- Observed Behaviour
- Outcome (Pass/Fail)
- Supporting Evidence

Supporting evidence may include console output, execution logs, screenshots, generated files, or equivalent artefacts that demonstrate the outcome of the test.

7. Submission Deliverables, Assessment and Submission Instructions

7.1 Submission Deliverables

Students shall submit the following deliverables through the course Moodle page.

D1 – Source Code

Submit the complete source code of the implementation.

D2 – Assignment 1 Report

The Assignment 1 Report shall be prepared as a well-organized technical report and shall include the following sections:

- Design Summary
- Testing Results (TR-1 to TR-8)
- Supporting Evidence
- Performance Analysis
- Discussion

The report shall clearly document the implementation assumptions, nonce management approach, Associated Data (AAD) selection, replay handling strategy, and observations made during testing and performance evaluation.

D3 – README

The README shall include:

- Software requirements
- External libraries used
- Build instructions
- Execution instructions

7.2 Assessment Criteria

Assessment Component	Weight
Positive Baseline Test (TR-1)	10%
Ciphertext Integrity Test (TR-2)	8%
Authentication Tag Test (TR-3)	8%
Associated Data (AAD) Test (TR-4)	8%
Replay Test (TR-5)	8%

Wrong-Key Test (TR-6)	8%
Nonce Management Verification (TR-7)	10%
Performance Evaluation (TR-8)	15%
Assignment 1 Report	15%
Individual Viva	10%
Total	100%

Note: Unless otherwise specified, every Testing Requirement shall be demonstrated separately using both supported AEAD configurations (AES-GCM and ChaCha20-Poly1305). Demonstrating a Testing Requirement using only one supported AEAD configuration will receive partial credit for that assessment component.

7.3 Submission Instructions

- All assignment submissions shall be uploaded through the course Moodle page.
- The submission deadline is August 21 2026..
- Late submissions: up to 24 hours – 20% deduction; between 24 and 48 hours – 40% deduction; beyond 48 hours – submission will not be accepted.
- No deadline extensions will be granted except under officially approved institute policies.
- Students are responsible for ensuring that the uploaded submission is complete and readable before the submission deadline.
- Every student shall attend the scheduled individual viva as part of the assignment evaluation.